Increasing exponential functions

1

a

x	-5	-4	-3	-2	-1	0	1	2	3	4	5	10
y	$\frac{1}{243}$	$\frac{1}{81}$	$\frac{1}{27}$	$\frac{1}{9}$	$\frac{1}{3}$	1	3	9	27	81	243	$59\,049$

b As x increases, the function increases at a faster and faster rate.

c All real x

d $y > 0$

2

a $\frac{1}{16}$

b 1

c 16

d None. For all values of x, it will have a positive value.

e It gets larger, approaching infinity.

f It gets smaller, approaching zero.

3

a Positive

b 0

c $y = 0$ is a horizontal asymptote of the curve.

4

a $(0, 1)$

b No

c All real x

d $y > 0$

e $y = 128$

f $x = 8$

5

a No. For $a > 0$ the expression a^x is greater than zero for all real x.

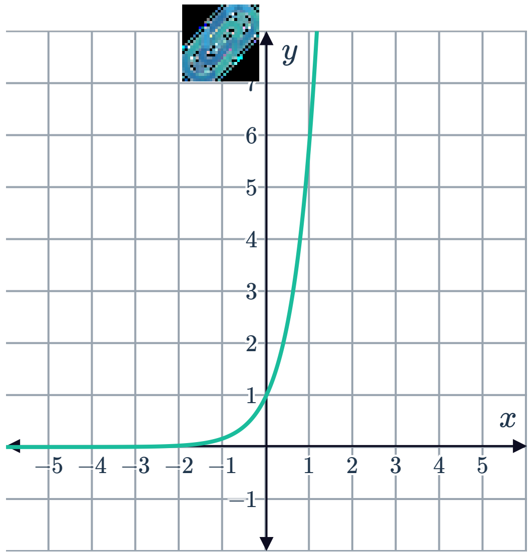
b ∞

c 0

d 1

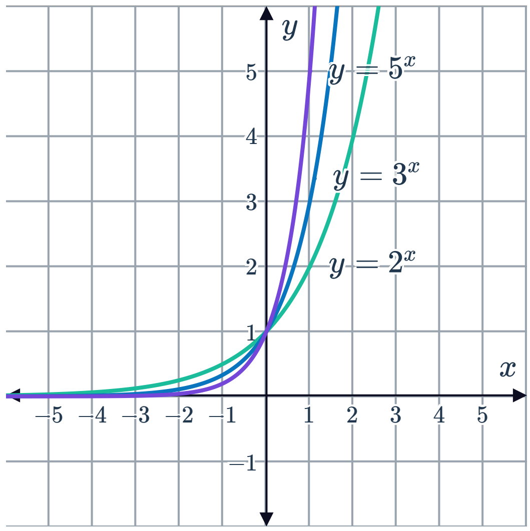
e 0

f



6

a



b Shared y-intercept of $(0, 1)$

c As x gets larger, the function increases at an increasing rate and the values approach ∞.

7 $m = 8$

8

a 125 **b** 1 **c** $\frac{1}{5}$ **d** -2

9 80 units

Decreasing exponential functions

10

a

x	-5	-4	-3	-2	-1	0	1	2	3	4	5	10
y	32	16	8	4	2	1	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{16}$	$\frac{1}{32}$	$\frac{1}{1024}$

b As x increases, the function decreases at a slower and slower rate.

c 1

d All real x

e $y > 0$

11

a As x increases, y increases at an increasing rate.

b As x increases, y decreases at a decreasing rate.

12

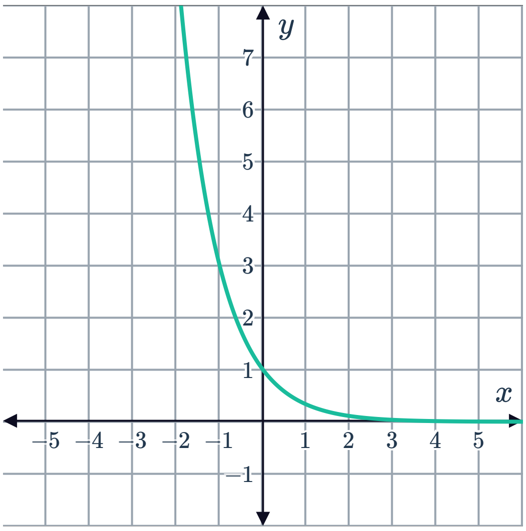
a 1

b

x	-3	-2	-1	0	1	2	3
y	27	9	3	1	$\frac{1}{3}$	$\frac{1}{9}$	$\frac{1}{27}$

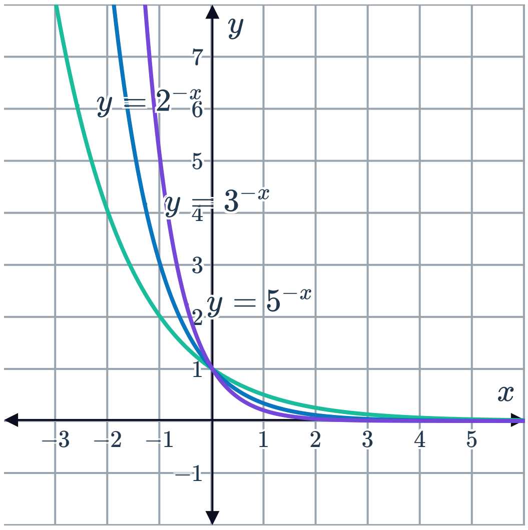
c $y = 0$

d



13

a



b Shared y-intercept of $(0, 1)$

c As x gets large, the function values approach 0.

14

a No. For $a > 0$ the expression a^x is greater than zero for all real x.

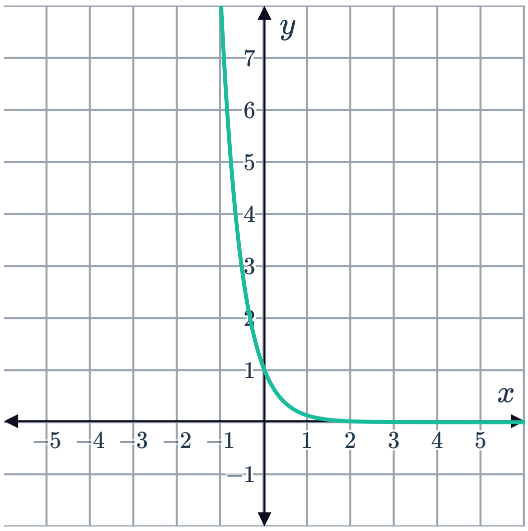
b 0

c ∞

d 1

e 0

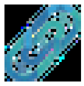
f



15

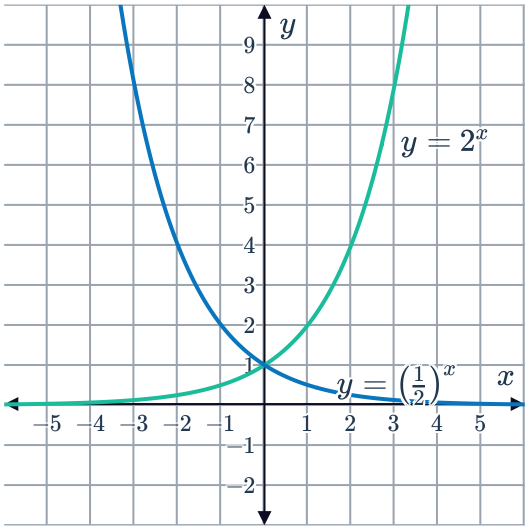
a

i Yes **ii** Yes **iii** No **iv** No



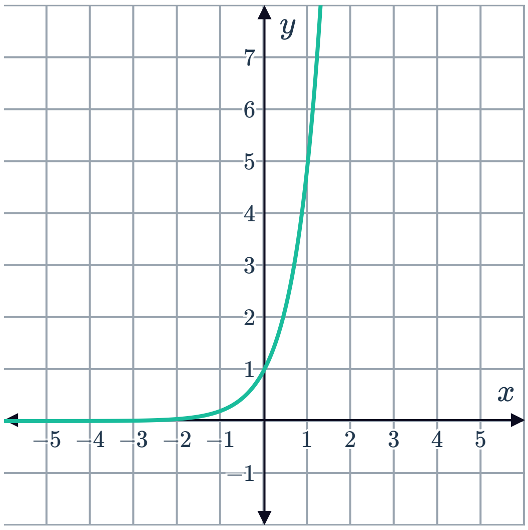
b Reflection about the y-axis.

c



16

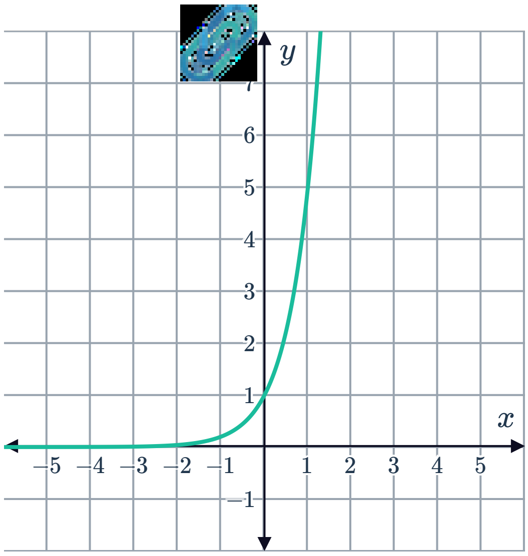
a



b

x	-3	-2	-1	0	1	2	3
y	$\frac{1}{125}$	$\frac{1}{25}$	$\frac{1}{5}$	1	5	25	125

c



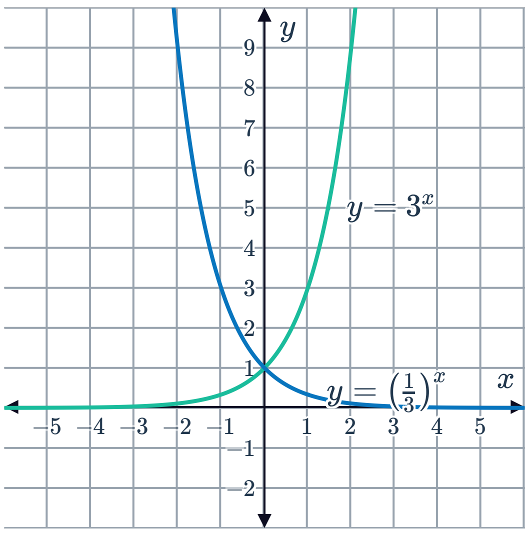
d The two graphs are the same.

17

a $y = 3^{-x}$

b Reflection about the y-axis.

c



18

a $(0, 1)$

b As x gets large $y = 2^x$ becomes large while $y = 2^{-x}$ approaches 0.

19

a Increasing

b Decreasing

c Decreasing

d Increasing

e Increasing

f Decreasing

g Decreasing

h Increasing

20 No, both functions are always greater than zero.

21

a 1

b -2

c 3

d 3

Evaluate exponential functions

22

a 5

b 80

c $\frac{5}{4}$

23 67

24

a 83

b $\frac{55}{27}$

25

a 1

b $\frac{9}{16}$

26

a $\frac{1}{8}$

b $\frac{1}{2^{\frac{11}{4}}}$

27

a $\frac{730}{27}$

b $\frac{730}{27}$

c Yes

28

a 5

b $\frac{1}{243}$

c 5

29 $50\,313.28$

30 2409.20

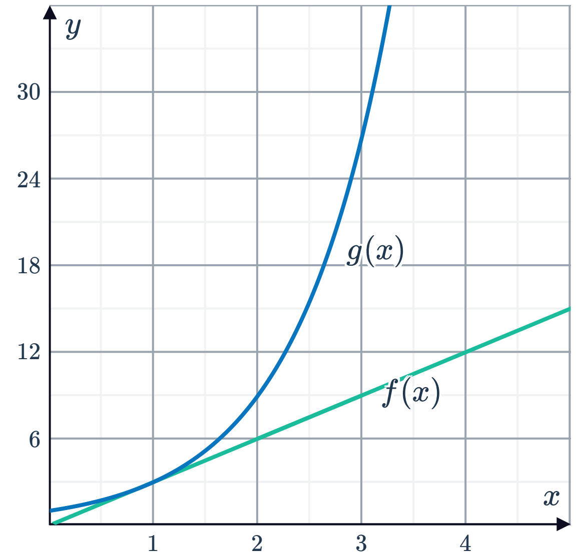
Compare growth

31

a

x	0	1	2	3	4
$f(x)$	0	3	6	9	12
$g(x)$	1	3	9	27	81

b



c

i 3 **ii** 3

d $g\left(x\right)=3^x$

e

i 24 **ii** 6561

f $f\left(x\right)=3x$ increases at a constant rate. $g\left(x\right)=3^x$ increases at an increasing rate.

32 $y=5^x$

33 $y=5^{-x}$

34

a Graph R

b For $x<0$, the graph of $y=6^x$ is below the graph of $y=4^x$. This is because negative values of x result in fractional function values, and for any negative value of x, 6^x will result in a smaller value than 4^x.

35 Ivan is incorrect. $g\left(x\right)>f\left(x\right)$ for some values of $x\geq 1$. However, as $f\left(x\right)$ is increasing at an increasing rate while $g\left(x\right)$ increases at a constant rate, at some point $f\left(x\right)$ will become greater in value.

36

a

n	1	2	3	4	5
$f(n)$	6	11	16	21	26
$g(n)$	6	12	24	48	96

b The consecutive terms of $f(n)$ are increasing linearly by 5, while the consecutive terms of $g(n)$ are increasing by a factor of 2.

c John is correct, since the second term of $g(n)$ is larger than $f(n)$, and the terms of $g(n)$ increase at a faster rate they will remain larger than the terms of $f(n)$.

37

a

x	4	5	6
$f(x)$	22	27	32
$g(x)$	405	1215	3645

b

i 5 **ii** 3 **iii** 5 **iv** 3

c The linear function increases by a constant difference and the exponential function increases by a constant multiplicative factor.

38

a 4 units **b** 2

c The exponential function

39

a Points F, B, D **b** Points E, C, A **c** 6 units **d** 2

40 Sophia

41

a

x	$f(x)$	Increase in $f(x)$	$g(x)$	Increase in $g(x)$
0	1	–	0	–
1	2	1	2	2
2	4	2	8	6
3	8	4	18	10
4	16	8	32	14
5	32	16	50	18
6	64	32	72	22
7	128	64	98	26

b The quadratic function increases more rapidly at first, but then the exponential function starts to increase more rapidly.

42

a The exponential function's increase over a 1 unit interval of x is greater than the quadratic function's increase over the same interval.

b The gap between the functions becomes greater and greater.